

A log-cosine integral and a pole on the contour

the hidden circle $|u - \frac{1}{2}| = \frac{1}{2}$ and the half-residue at $u = 1$

$$\boxed{\int_0^{\pi/2} \frac{\log \cos x}{x^2 + \log^2 \cos x} dx = \frac{\pi}{2} \left(1 - \frac{1}{\log 2}\right)}.$$

Overview. The quadratic form $x^2 + \log^2 \cos x$ in the denominator is the squared modulus of the complex number

$$w(x) = \log \cos x + ix,$$

and the numerator is its real part, so the integrand is $\operatorname{Re} \frac{1}{w(x)}$. Exponentiating,

$$u = e^{w(x)} = \cos x e^{ix} = \frac{1 + e^{2ix}}{2},$$

which is a circle of radius $\frac{1}{2}$ centred at $\frac{1}{2}$, traversed once as x runs over $(-\frac{\pi}{2}, \frac{\pi}{2})$. The integral therefore becomes a contour integral of

$$H(u) = \frac{1}{(u - \frac{1}{2}) \operatorname{Log} u}$$

around that circle. Two singular points matter: an interior pole at $u = \frac{1}{2}$, the centre, which produces the term $-\frac{\pi}{2 \log 2}$, and a pole *on* the contour at $u = 1$, which produces the term $\frac{\pi}{2}$ through the half-residue rule. The endpoint $u = 0$, where $\log \cos x$ blows up, contributes nothing.

The tangency structure here is the mirror image of the log sin case: there the circle is $|u - \frac{i}{2}| = \frac{1}{2}$ and the mean-value property applies cleanly; here the circle is $|u - \frac{1}{2}| = \frac{1}{2}$ and a pole sits on it, so the mean-value property must be replaced by a principal-value residue computation.

1. The hidden complex variable

Set

$$w(x) = \log \cos x + ix, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}.$$

Then

$$\frac{1}{w(x)} = \frac{\overline{w(x)}}{|w(x)|^2} = \frac{\log \cos x - ix}{\log^2 \cos x + x^2},$$

so the integrand of the problem is exactly

$$\operatorname{Re} \frac{1}{w(x)} = \frac{\log \cos x}{x^2 + \log^2 \cos x}. \quad (1)$$

Since $\cos$ is even, $w(-x) = \overline{w(x)}$, hence $\frac{1}{w(-x)} = \frac{1}{\overline{w(x)}}$, and the imaginary parts cancel on a symmetric interval:

$$J := \operatorname{PV}_{x=0} \int_{-\pi/2}^{\pi/2} \frac{dx}{w(x)} = 2 \int_0^{\pi/2} \operatorname{Re} \frac{1}{w(x)} dx = 2I. \quad (2)$$

Why a principal value, and why only for J . At $x = 0$ we have $w(0) = 0$, so $1/w$ is singular there. Expanding,

$$w(x) = ix - \frac{x^2}{2} + O(x^4), \quad \frac{1}{w(x)} = -\frac{i}{x} + \frac{1}{2} + O(x),$$

so the singular part is purely imaginary and odd; the principal value at $x = 0$ exists and kills it. The real part is bounded:

$$\operatorname{Re} \frac{1}{w(x)} \longrightarrow -\frac{1}{2}, \quad x \rightarrow 0,$$

so the original integral I converges as an ordinary integral. At the other endpoint $x \rightarrow \frac{\pi}{2}$, $\log \cos x \rightarrow -\infty$ and the integrand behaves like $1/\log \cos x \rightarrow 0$. Both endpoints are harmless for I .

2. Exponentiating: the circle $|u - \frac{1}{2}| = \frac{1}{2}$

Put $u = e^{w(x)}$. Then

$$u = \cos x e^{ix} = \frac{e^{-ix} + e^{ix}}{2} e^{ix} = \frac{1 + e^{2ix}}{2},$$

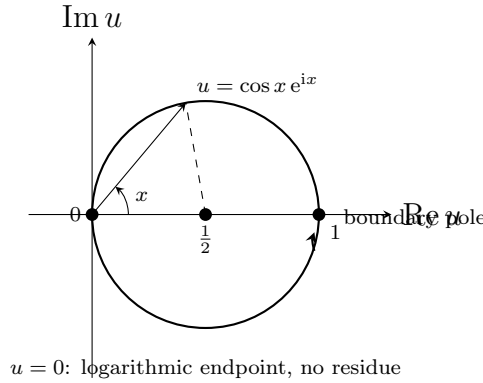
that is,

$$\boxed{u - \frac{1}{2} = \frac{1}{2} e^{2ix}}. \quad (3)$$

As x runs from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$, the angle $2x$ runs from $-\pi$ to π , so u traverses

$$C : \left| u - \frac{1}{2} \right| = \frac{1}{2}$$

exactly once, positively oriented. The circle passes through $u = 0$ (at $x = \pm\frac{\pi}{2}$) and $u = 1$ (at $x = 0$).



The branch is fixed by the geometry: every point of C except $u = 0$ has $\operatorname{Re} u > 0$, so the correct logarithm is the principal one restricted to the right half-plane,

$$\operatorname{Log} u = \log |u| + i \operatorname{Arg} u, \quad -\frac{\pi}{2} < \operatorname{Arg} u < \frac{\pi}{2},$$

and with this choice $w(x) = \operatorname{Log} u$ identically, since $\operatorname{Im} w = x \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

Differentiating (3),

$$du = i e^{2ix} dx = 2i \left(u - \frac{1}{2} \right) dx, \quad dx = \frac{du}{2i \left(u - \frac{1}{2} \right)}. \quad (4)$$

Substituting (4) into (2),

$$J = \frac{1}{2i} \operatorname{PV} \int_C \frac{du}{\left(u - \frac{1}{2} \right) \operatorname{Log} u} = \frac{1}{2i} \operatorname{PV} \int_C H(u) du, \quad H(u) = \frac{1}{\left(u - \frac{1}{2} \right) \operatorname{Log} u}. \quad (5)$$

3. The three special points of H on and inside C

(a) $u = \frac{1}{2}$, **interior, simple pole**. The centre of the circle is inside C and $\operatorname{Log} \frac{1}{2} = -\log 2 \neq 0$, so

$$\operatorname{Res}_{u=1/2} H = \frac{1}{\operatorname{Log} \frac{1}{2}} = -\frac{1}{\log 2}. \quad (6)$$

(b) $u = 1$, **on the contour, simple pole**. Here $\operatorname{Log} u$ vanishes to first order: $\operatorname{Log} u = (u - 1) - \frac{(u-1)^2}{2} + \dots$. Hence H has a simple pole at $u = 1$ with

$$\operatorname{Res}_{u=1} H = \lim_{u \rightarrow 1} \frac{u - 1}{\left(u - \frac{1}{2} \right) \operatorname{Log} u} = \frac{1}{\frac{1}{2}} = 2. \quad (7)$$

This is the point $x = 0$, the same singularity for which the principal value was introduced in Section 1.

(c) $u = 0$, **on the contour, not a pole**. Although $\text{Log } u$ is singular at the origin, H is not: $\frac{1}{\text{Log } u} \rightarrow 0$ as $u \rightarrow 0$. If the contour is indented by a small arc of radius ε at the origin, the contribution is bounded by

$$\frac{C\varepsilon}{|\log \varepsilon|} \rightarrow 0,$$

since the arclength is $O(\varepsilon)$, the factor $(u - \frac{1}{2})^{-1}$ is bounded there, and $|\text{Log } u| \geq |\log \varepsilon|$. So the origin may be ignored.

4. Principal value with a pole on the contour

Lemma (half-residue). Let C be a positively oriented smooth closed contour, and let H be holomorphic near C and inside it except for finitely many poles, those on C being simple. Then

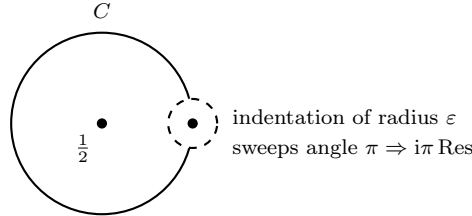
$$\boxed{\text{PV} \int_C H(u) du = 2\pi i \sum_{a \in \text{Int } C} \text{Res}_{u=a} H + i\pi \sum_{a \in C} \text{Res}_{u=a} H.} \quad (8)$$

Sketch. Indent C at each boundary pole a by a small circular arc of radius ε pushed outside the region. The indented contour encloses only the interior poles, giving the first sum. A simple pole with residue r contributes, over an arc of opening angle π (the tangent line, in the limit $\varepsilon \rightarrow 0$),

$$\int_{\text{arc}} \frac{r}{u-a} du = r i \int d\varphi = i\pi r$$

with sign fixed by the orientation, and the remaining regular part of H contributes $O(\varepsilon)$. Passing to the limit gives (8). $\square$

The geometric reading is that a pole sitting on a smooth boundary is “half inside”, so it enters with half its usual weight $2\pi i \rightarrow i\pi$.



5. Evaluation

Applying (8) with the residues (6) and (7), and with the origin excluded by Section 3(c),

$$\text{PV} \int_C H(u) du = 2\pi i \left(-\frac{1}{\log 2} \right) + i\pi \cdot 2 = 2\pi i \left(1 - \frac{1}{\log 2} \right).$$

By (5),

$$J = \frac{1}{2i} \cdot 2\pi i \left(1 - \frac{1}{\log 2} \right) = \pi \left(1 - \frac{1}{\log 2} \right),$$

and by (2), $I = \frac{J}{2}$:

$$\boxed{\int_0^{\pi/2} \frac{\log \cos x}{x^2 + \log^2 \cos x} dx = \frac{\pi}{2} \left(1 - \frac{1}{\log 2} \right) = -0.69538374 \dots}$$

The sign is a useful check: $\log \cos x < 0$ on $(0, \frac{\pi}{2})$, so the integrand is negative throughout, and indeed $\frac{1}{\log 2} > 1$. Numerical quadrature gives $-0.6953837441187 \dots$, agreeing to all displayed digits.

6. The anatomy of the answer

The two terms of the result come from two geometrically distinct sources, and it is worth separating them.

Point	Nature	Contribution to I	Source
$u = \frac{1}{2}$	interior simple pole	$-\frac{\pi}{2 \log 2}$	The centre of the circle: this is the mean-value term, the analogue of $\pi F(w_0)$.
$u = 1$	simple pole on C	$+\frac{\pi}{2}$	The point $x = 0$, where $w(0) = 0$; a half-residue.
$u = 0$	logarithmic endpoint	0	$1/\text{Log } u \rightarrow 0$ faster than the arclength shrinks.

Written as $I = \frac{\pi}{2} - \frac{\pi}{2 \log 2}$, the first term is pure boundary geometry and carries no arithmetic constant, whereas the $\log 2$ is the value $\text{Log } \frac{1}{2}$ at the centre of the circle. This is why $\log 2$ appears in a problem whose statement mentions no logarithm of a constant: $\frac{1}{2}$ is the centre of the hidden circle.

7. Comparison with the mean-value case

For the $\log \sin$ family one has $u = \sin x e^{ix}$, the circle $|u - \frac{i}{2}| = \frac{1}{2}$ tangent to the origin, and for holomorphic F the clean identity

$$\int_0^\pi F(\log \sin x + ix) dx = \pi F\left(\text{Log } \frac{i}{2}\right),$$

the mean-value property transported by $u \mapsto \text{Log } u$. The temptation is to write the present problem the same way with $F(w) = 1/w$ and conclude $\int \frac{dx}{w} = \pi/w_0$ with $w_0 = \text{Log } \frac{1}{2} = -\log 2$, giving $-\frac{\pi}{\log 2}$ for J , which is wrong by exactly π .

The discrepancy is the whole content of the problem: $F(w) = 1/w$ is *not* holomorphic on the closed region, because $w_0 = 0$ is attained on the boundary curve at $x = 0$. In u coordinates that is the pole of H at $u = 1 \in C$. The mean-value property applies to the regular part and yields $-\frac{\pi}{\log 2}$; the half-residue at the boundary pole restores the missing $+\pi$.

	log sin family	this problem
Curve	$u = \sin x e^{ix}$	$u = \cos x e^{ix}$
Circle	$ u - \frac{i}{2} = \frac{1}{2}$	$ u - \frac{1}{2} = \frac{1}{2}$
Branch strip	$0 < \text{Im } w < \pi$	$ \text{Im } w < \frac{\pi}{2}$
Centre	$w_0 = -\log 2 + \frac{\pi i}{2}$	$w_0 = -\log 2$
Obstruction	none for $F = \text{Log } \text{Log}$	$F = 1/w$ has a pole at $w'_0 = 0 \in \partial$
Result	$\pi F(w_0)$	$\pi F(w_0) + i\pi \cdot \frac{1}{2i} \text{Res}$

8. Variants that come free

Once (5) is set up, other integrands are simply other H .

The imaginary part. Taking $-\operatorname{Im} \frac{1}{w}$ instead of $\operatorname{Re} \frac{1}{w}$ gives an odd integrand, so

$$\int_0^{\pi/2} \frac{x}{x^2 + \log^2 \cos x} dx$$

is *not* obtained by this symmetry argument. It corresponds to the divergent part removed by the principal value, and indeed diverges logarithmically at $x = 0$, where the integrand behaves like $1/x$.

Shifted denominators. For real $a > 0$, replace w by $w + a$, i.e. take

$$H_a(u) = \frac{1}{\left(u - \frac{1}{2}\right) (\operatorname{Log} u + a)}.$$

The zero of $\operatorname{Log} u + a$ is $u = e^{-a} \in (0, 1)$, which satisfies $|e^{-a} - \frac{1}{2}| < \frac{1}{2}$ and hence lies strictly *inside* C . So for $a > 0$ there is no pole on the contour, (8) reduces to the ordinary residue theorem with the two interior poles $u = \frac{1}{2}$ and $u = e^{-a}$, and

$$\operatorname{Res}_{u=1/2} H_a = \frac{1}{a - \log 2}, \quad \operatorname{Res}_{u=e^{-a}} H_a = \frac{1}{\left(u - \frac{1}{2}\right) \frac{1}{u}} \Big|_{u=e^{-a}} = \frac{e^{-a}}{e^{-a} - \frac{1}{2}}.$$

This gives, for every $a > 0$,

$$\boxed{\int_0^{\pi/2} \frac{\log \cos x + a}{(\log \cos x + a)^2 + x^2} dx = \frac{\pi}{2} \left[\frac{1}{a - \log 2} + \frac{e^{-a}}{e^{-a} - \frac{1}{2}} \right]}. \quad (9)$$

Formula (9) is discontinuous at $a = 0$. Letting $a \rightarrow 0^+$ in (9),

$$\frac{\pi}{2} \left[-\frac{1}{\log 2} + \frac{1}{1 - \frac{1}{2}} \right] = \frac{\pi}{2} \left(2 - \frac{1}{\log 2} \right),$$

whereas the value at $a = 0$ computed in Section 5 is $\frac{\pi}{2} \left(1 - \frac{1}{\log 2} \right)$. The jump is exactly $\frac{\pi}{2}$, half the interior contribution of the pole that is crossing the contour.

The same jump is visible on the real side without any complex analysis. Near $x = 0$ we have $\log \cos x = -\frac{x^2}{2} + O(x^4)$, so for small $a > 0$ the integrand of (9) is approximately $\frac{a}{a^2 + x^2}$, whose integral over a fixed neighbourhood of 0 is $\arctan \frac{\delta}{a} \rightarrow \frac{\pi}{2}$. A spike of height $1/a$ and width a carries a fixed mass $\frac{\pi}{2}$ and disappears at $a = 0$, where the integrand is merely bounded, tending to $-\frac{1}{2}$. The half-residue rule of Section 4 is the complex-analytic bookkeeping for precisely this Poisson-kernel spike.